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Keishibukuryogan ameliorates glucose intolerance and hyperlipidemia in Otsuka Long-Evans Tokushima Fatty (OLETF) rats

Takako Nakagawa^{a,*}, Hirozo Goto^{b,c}, Ghazi Hussein^d, Hiroaki Hikiami^b,
Naotoshi Shibahara^{a,c}, Yutaka Shimada^{b,c}

^a Department of Kampo Diagnostics, Institute of Natural Medicine, University of Toyama, 2630 Sugitani, Toyama 930-0194, Japan

^b Department of Japanese Oriental Medicine, Faculty of Medicine, University of Toyama, 2630 Sugitani, Toyama 930-0194, Japan

^c 21st Century COE Program, University of Toyama, 2630 Sugitani, Toyama 930-0194, Japan

^d Division of Medicinal Pharmacology, Institute of Natural Medicine, University of Toyama, 2630 Sugitani, Toyama 930-0194, Japan

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ABSTRACT

Keishibukuryogan, one of the traditional herbal formulations, is used clinically to improve blood circulation. In this study, we examined the effects of keishibukuryogan on glucose and lipids metabolism in Otsuka Long-Evans Tokushima Fatty (OLETF) rats, an animal model of type 2 diabetes. Forty-five-week-old male OLETF rats were divided into three groups: diabetic control rats given a standard chow; diabetic rats given keishibukuryogan (3%, w/w in chow); diabetic rats given pioglitazone (0.01%, w/w in chow). Oral administration of keishibukuryogan produced significant improvement against impaired glucose tolerance. On the other hand, fasting serum glucose and insulin levels, and the homeostasis index of insulin resistance did not change by keishibukuryogan treatment. Against lipid parameters, keishibukuryogan significantly lowered serum total cholesterol and triglyceride levels, and the hepatic total cholesterol level. Keishibukuryogan treatment also significantly reduced the serum leptin level, but it had no effect on the serum adiponectin level. Additionally, keishibukuryogan showed significant effects on epididymal adipose tissue by decreasing the size of fat cells and on skeletal muscle by reducing TNF- α protein content. From these results, it was suggested that keishibukuryogan exerts beneficial effects on the features associated with type 2 diabetes.

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1. Introduction

Diabetes mellitus is a serious worldwide health problem. It is a matter of concern that global prevalence of diabetes in adults is estimated to rise to 300 million cases, and from the view point of age profile, the majority of people with diabetes will be over 65 years old in the year 2025 [1]. Therefore, it is important to seek effective therapeutic interventions for the prevention

and treatment of diabetes and its complications, and the improvement of the quality of life of diabetic patients. Under these circumstances, the use of alternative medicines including traditional herbal medicines has increasingly become the focus of attention.

Traditional Chinese/Japanese (Kampo) medicine has a long history and has contributed to the prevention and treatment of various diseases, including many subjective symptoms of

* Corresponding author. Tel.: +81 76 434 7393; fax: +81 76 434 0366.

E-mail address: takako26@med.u-toyama.ac.jp (T. Nakagawa).

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diabetic patients. Some studies have examined the effects of traditional Chinese/Japanese medicine on insulin resistance, using STZ-induced diabetic rats [2,3] and KK-A^y mice [4]. In addition, there are case reports regarding a favorable effect on insulin resistance and blood glucose [5,6].

Keishibukuryogan, one of the prescriptions in Japanese traditional medicines, has been widely administered to patients with blood stagnation for improving blood circulation. In this connection, Itoh et al. reported that keishibukuryogan improves blood viscosity and conjunctival microcirculation in patients with cerebro-spinal vascular diseases [7]. In animal experiments, keishibukuryogan was reported to prevent the progression of atherosclerosis [8], and to preserve vascular endothelial function in cholesterol-fed rabbits [9] and in hypertensive rats [10]. Furthermore, in diabetic rats, it was demonstrated to have protective effects against vascular injury [11], and to delay the development of diabetic kidney disease [12,13].

In this study, we investigated the effects of keishibukuryogan on the features associated with type 2 diabetes, the most common form of the disease. In addition, to evaluate the efficacy of this herbal medicine, we employed pioglitazone (insulin-sensitizing agent) as a positive control.

2. Materials and methods

2.1. Test drug

Powdered keishibukuryogan was purchased from Uchida Wakanyaku (Tokyo, Japan). It is composed of equal parts, by weights, of the following five crude drugs: Cinnamon Cortex, *Cinnamomum cassia* Blume (Lauraceae); Poria, *Poria cocos* Wolf (Polyporaceae); Moutan Cortex, *Paeonia suffruticosa* Andrews (Paeoniaceae); Persicae Semen, *Prunus persica* Batsch (Rosaceae) and Paeoniae Radix, *Paeonia lactiflora* Pallas (Paeonia-

ceae). Fig. 1 shows the three-dimensional HPLC chart of keishibukuryogan. The HPLC conditions were as follows: keishibukuryogan extract (2.5 g) was obtained with 20 mL of methanol under ultrasonication for 30 min. The solution was filtered with membrane filter (0.45 μ m) and then subjected to HPLC analysis. HPLC equipment was controlled with an SLC-10A (Shimadzu, Tokyo, Japan) using a TSK-GEL ODS-80TS column (4.6 ψ $\times$ 250 mm), eluting with solvents (A) 0.05 M AcONH₄ (pH 3.6) and (B) CH₃CN. A linear gradient of 100% A and 0% B, changing over 60 min to 0% A and 100% B, was used. The flow rate was controlled with an LC-10AD pump at 1.0 mL/min. The eluate from the column was monitored, and three-dimensional data were processed by SPD-M10A diode array detectors. All assigned peaks were identified by co-injection test with authentic samples and compared with UV spectral data.

2.2. Animals and drug treatments

Male Otsuka Long-Evans Tokushima Fatty (OLETF) rats were kindly supplied by the Tokushima Research Institute, Otsuka Pharmaceutical Company (Tokushima, Japan). The animals were fed standard laboratory rat chow, allowed to drink tap water ad libitum, and were housed in an automatically controlled room (temperature about 23 °C and humidity about 60%) with a conventional light/dark cycle. At the age of 45 weeks, male OLETF rats were divided into three groups (eight rats each) based on body weight and plasma glucose. The control group was fed a standard chow, and keishibukuryogan and pioglitazone groups were fed a standard chow containing keishibukuryogan (3%, w/w) and pioglitazone (0.01%, w/w), respectively. During the experimental period, consumption of diet was kept at the same amount for all animals (30 g/(rat/day)). After 6 weeks of treatment, the rats were sacrificed and blood samples were obtained from each rat. Tissues were quickly frozen and kept at –80 °C until analysis.

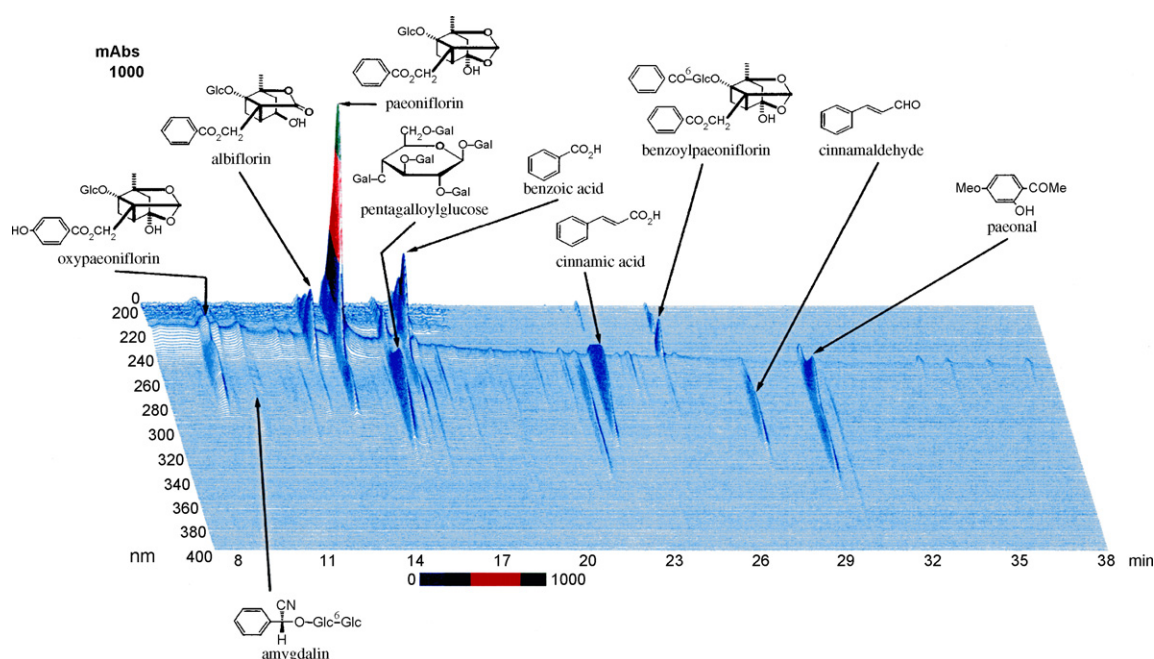


Fig. 1 – Three-dimensional HPLC profile of keishibukuryogan. Glc: β -D-glucopyranosyl; Gal: galloyl.

Table 1 – Effects of keishibukuryogan and pioglitazone on body and tissue weight

	Body weight (g)			Liver weight (g/100 g BW)	Kidney weight (g/100 g BW)	Epididymal adipose tissue (g/100 g BW)
	Week 0	Week 3	Week 6			
Control	580 ± 19	596 ± 18	600 ± 20	2.93 ± 0.09	0.707 ± 0.022	2.86 ± 0.28
Keishibukuryogan	588 ± 19	612 ± 27	612 ± 24	2.79 ± 0.09	0.711 ± 0.023	2.34 ± 0.12
Pioglitazone	589 ± 22	609 ± 25	622 ± 25	2.42 ± 0.04**	0.656 ± 0.021	2.76 ± 0.18

Data are presented as mean ± S.E. ** $p < 0.01$ vs. OLETF control rats.

All experimental procedures were performed in accordance with the standards established by the 'Guide for the Care and Use of Laboratory Animals at the University of Toyama'.

2.3. Analysis of blood samples

Serum levels of glucose, triglyceride, total cholesterol and free fatty acid were determined using commercial kits (Glucose CII-Test Wako, Triglyceride E-Test Wako, Cholesterol E-Test Wako, NEFA C-Test Wako, obtained from Wako Pure Chemical Industries Ltd., Osaka, Japan). Serum insulin, leptin and adiponectin were determined by commercial ELISA kit (Insulin ELISA kit, leptin/rat ELISA kit, Morinaga, Kanagawa, Japan; mouse/rat adiponectin ELISA kit, Otsuka Pharmaceuticals, Tokyo, Japan). Homeostasis model assessment of insulin resistance (HOMA-IR) was performed using the following formula: $\text{HOMA-IR} = [\text{fasting insulin } (\mu\text{U/ml}) \times \text{fasting glucose (mM)}] / 22.5$ [14].

2.4. Oral glucose tolerance test (OGTT)

OGTT was carried out after 3 and 5 weeks of treatment. After 16 h fasting, glucose solution (2 g/kg body weight) was orally administered, and blood samples were drawn at 0, 30, 60, and 120 min post-administration from the tail vein. Blood glucose levels were determined using an Antsense III (Horiba, Tokyo, Japan).

2.5. Determination of total cholesterol and triglyceride levels in liver

The liver of each rat was homogenized, lipids were extracted with a mixture of chloroform and methanol (2:1, v/v), and the amounts of total cholesterol and triglyceride levels were determined using the Wako kit described above.

2.6. Measurement of TNF- α content in skeletal muscle

Tissue was homogenized in a 100 mM Tris-HCl buffer (pH 7.6) containing 1 M NaCl, 2% fat-free bovine serum albumin, 2 mM EDTA, 80 units/L aprotinin, and 0.02% NaN_3 . The aliquots were centrifuged at $13,500 \times g$ for 30 min, and the supernatants were subjected to a commercial ELISA assay kit (rat TNF- α , UltraSensitive, BioSource International, Inc., Camarillo, CA).

2.7. Histological study

pididymal adipose tissue was fixed with formalin (10%), embedded in paraffin, and sections were stained with

hematoxylin/eosin. To examine the size of white adipocytes, the number of adipocytes was counted in a limited area (capture size, 640×480 pixels, equivalent to $12,288 \mu\text{m}^2$) of each stained specimen by light microscopy (Olympus Provis AX80, Olympus Optical Co. Ltd., Tokyo).

2.8. Statistical analysis

Values are presented as mean ± S.E. Differences among groups were analyzed by Dunnett's test and those at $p < 0.05$ were accepted as significant.

3. Results

3.1. Body and tissue weights

Changes in body, liver, kidney and epididymal adipose tissue weights are summarized in Table 1. During the experimental period, body weight did not change among the three groups, i.e., control, keishibukuryogan and pioglitazone groups. On the other hand, liver weight of the pioglitazone group was significantly lower than that of the control group. Compared with the control group, kidney and epididymal adipose tissue weights did not show significant changes in either the keishibukuryogan or pioglitazone group.

Table 2 – Effects of 3- and 6-week treatment of keishibukuryogan and pioglitazone on serum glucose and insulin levels, and HOMA-IR

	Duration of treatment		
	Week 0	Week 3	Week 6
Glucose (mg/dl)			
Control	138.8 ± 5.9	140.4 ± 6.9	133.0 ± 6.6
Keishibukuryogan	130.3 ± 7.4	132.6 ± 7.2	126.7 ± 5.7
Pioglitazone	137.5 ± 6.6	111.4 ± 3.1**	105.8 ± 9.5**
Insulin (pg/ml)			
Control	1070 ± 275	1101 ± 252	1261 ± 154
Keishibukuryogan	1100 ± 208	1270 ± 363	1082 ± 220
Pioglitazone	1020 ± 223	623 ± 84*	911 ± 18
HOMA-IR			
Control	6.75 ± 2.02	7.97 ± 1.89	7.62 ± 0.88
Keishibukuryogan	6.60 ± 1.10	7.74 ± 2.27	6.17 ± 1.52
Pioglitazone	5.75 ± 1.51	3.19 ± 0.51*	4.46 ± 0.13*

Data are presented as mean ± S.E. * $p < 0.05$, ** $p < 0.01$ vs. OLETF control rats.

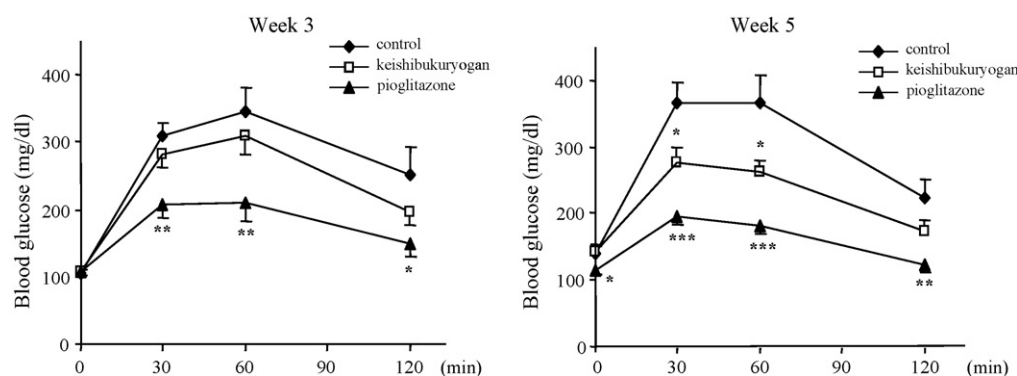


Fig. 2 – Oral glucose tolerance test after 3- and 5-week treatment of keishibukuryogan and pioglitazone. Each column represents mean $\pm$ S.E. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. OLETF control rats.

3.2. Glucose, insulin and HOMA-IR

Table 2 shows the effects of keishibukuryogan and pioglitazone on glucose and insulin levels, and HOMA-IR. Compared with the control group, the pioglitazone group had significantly decreased serum glucose level and HOMA-IR at 3 weeks, and this effect was retained at 6 weeks. The serum insulin level in the pioglitazone group was significantly lower at 3 weeks, and at 6 weeks it was also lower, but without significance. On the other hand, there were no significant changes in these parameters between the control and keishibukuryogan groups.

3.3. OGTT

Fig. 2 shows the effects of keishibukuryogan and pioglitazone on blood glucose levels after oral glucose load. Compared with the control group, pioglitazone significantly reduced blood glucose at 30, 60 and 120 min after glucose load in both OGTT, at 3 and at 5 weeks. Treatment of keishibukuryogan showed a tendency to decrease blood glucose at 3 weeks, and at 5 weeks

of treatment significant reductions were seen at 30 and 60 min compared to the control group.

3.4. Lipid metabolism

Table 3 shows the effects of each medicine on serum total cholesterol, triglyceride and free fatty acid levels after 3- and 6-week treatment. After 3-week treatment, serum total cholesterol was significantly reduced in both treated groups compared with control, and this effect remained after 6 weeks of treatment. The pioglitazone group showed strong reduction in the serum triglyceride level during the experimental period. The keishibukuryogan group also showed a significant decrease in serum triglycerides at 3 weeks, and a further reduction was observed at 6 weeks. There were no changes between the control and keishibukuryogan groups in serum free fatty acid levels, whereas the pioglitazone group showed significant decreases at 3 and 6 weeks compared with the control group. As shown in Fig. 3, the hepatic cholesterol level decreased significantly in both treated groups compared with the control group. Similarly, the hepatic triglyceride level

Table 3 – Effects of 3- and 6-week treatment of keishibukuryogan and pioglitazone on serum total cholesterol, triglyceride and free fatty acid levels

	Duration of treatment		
	Week 0	Week 3	Week 6
Total cholesterol (mg/dl)			
Control	146.3 $\pm$ 5.7	159.4 $\pm$ 8.8	167.0 $\pm$ 12.7
Keishibukuryogan	129.9 $\pm$ 8.4	137.2 $\pm$ 9.3*	128.6 $\pm$ 5.6*
Pioglitazone	138.6 $\pm$ 7.4	123.8 $\pm$ 6.0**	118.6 $\pm$ 11.2**
Triglyceride (mg/dl)			
Control	157.1 $\pm$ 9.5	250.0 $\pm$ 34.8	180.2 $\pm$ 34.3
Keishibukuryogan	153.7 $\pm$ 26.3	168.8 $\pm$ 18.0*	95.0 $\pm$ 7.8*
Pioglitazone	158.8 $\pm$ 15.7	64.0 $\pm$ 6.3**	65.5 $\pm$ 16.5**
Free fatty acid (mequiv./l)			
Control	0.911 $\pm$ 0.069	0.914 $\pm$ 0.062	1.062 $\pm$ 0.041
Keishibukuryogan	0.849 $\pm$ 0.047	0.796 $\pm$ 0.042	0.981 $\pm$ 0.072
Pioglitazone	0.911 $\pm$ 0.067	0.740 $\pm$ 0.045*	0.881 $\pm$ 0.069*

Data are presented as mean $\pm$ S.E. * $p < 0.05$, ** $p < 0.01$ vs. OLETF control rats.

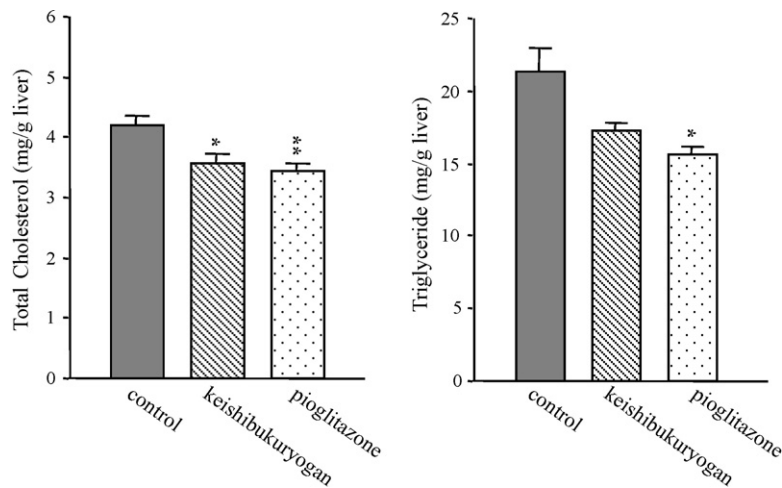


Fig. 3 – Effects of 6-week treatment of keishibukuryogan and pioglitazone on hepatic total cholesterol and triglyceride levels. Each column represents mean $\pm$ S.E. * $p < 0.05$, ** $p < 0.01$ vs. OLETF control rats.

in the pioglitazone group was significantly reduced, whereas the keishibukuryogan group showed a tendency to decrease without statistical significance.

3.5. Serum adiponectin and leptin

As shown in Table 4, the serum adiponectin level increased significantly in the pioglitazone group compared with the control group during the experimental period. There were no changes in adiponectin level between the control and keishibukuryogan groups. On the other hand, significant reduction in the serum leptin level was observed with 6-week keishibukuryogan treatment, as well as with 3- and 6-week pioglitazone treatment, compared with the control group.

3.6. TNF- α content

As shown in Fig. 4, the TNF- α protein content in skeletal muscle was significantly lowered in the keishibukuryogan and pioglitazone groups compared with control.

3.7. Histological examination

In microscopic studies, the number of adipocytes was significantly increased in the keishibukuryogan and pioglitazone groups compared with the control group, reflecting aggregates of small fat cells (Fig. 5).

4. Discussion

OLETF rats, a genetic model for the spontaneous development of type 2 diabetes, showed obesity, insulin resistance, glucose intolerance and dyslipidemia [15,16]. In this study, to evaluate

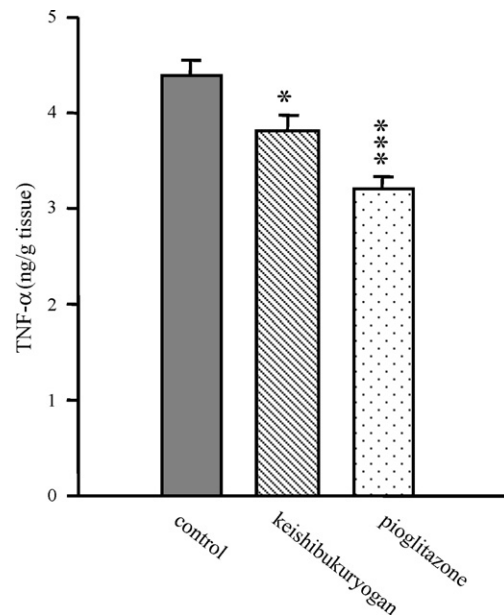


Fig. 4 – Effects of 6-week treatment of keishibukuryogan and pioglitazone on TNF- α levels in skeletal muscle. Each column represents mean $\pm$ S.E. * $p < 0.05$, * $p < 0.001$ vs. OLETF control rats.**

Table 4 – Effects of 3- and 6-week treatment of keishibukuryogan and pioglitazone on serum adiponectin and leptin levels

	Duration of treatment		
	Week 0	Week 3	Week 6
Adiponectin (μ g/ml)			
Control	4.29 $\pm$ 0.20	4.10 $\pm$ 0.14	4.06 $\pm$ 0.11
Keishibukuryogan	4.01 $\pm$ 0.25	4.01 $\pm$ 0.18	4.14 $\pm$ 0.22
Pioglitazone	3.80 $\pm$ 0.13	7.94 $\pm$ 0.36**	8.21 $\pm$ 0.42**
Leptin (ng/ml)			
Control	6.73 $\pm$ 1.24	7.78 $\pm$ 1.96	7.46 $\pm$ 1.75
Keishibukuryogan	5.84 $\pm$ 1.05	5.62 $\pm$ 1.28	4.50 $\pm$ 0.74*
Pioglitazone	5.60 $\pm$ 1.30	3.23 $\pm$ 1.05*	3.87 $\pm$ 0.78*

Data are presented as mean $\pm$ S.E. * $p < 0.05$, ** $p < 0.01$ vs. OLETF control rats.

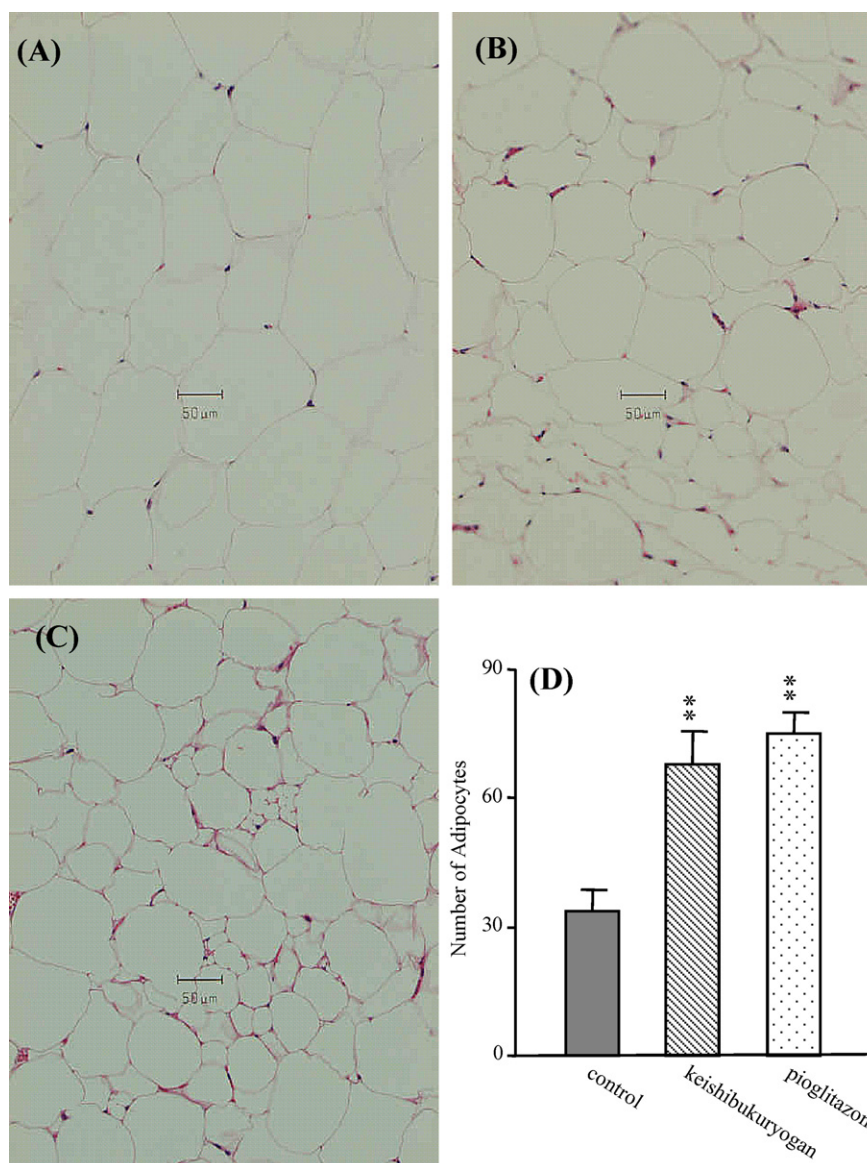


Fig. 5 – Representative morphology of epididymal white adipose tissue: (A) control, (B) keishibukuryogan, (C) pioglitazone, and (D) the number of adipocytes after 6-week treatment. Each column represents mean $\pm$ S.E. ** $p < 0.01$ vs. OLETF control rats.

the efficacy of keishibukuryogan, we employed the insulin-sensitizing agent pioglitazone as positive control. As also reported by several studies [17–19], pioglitazone remarkably improved hyperglycemia, hyperlipidemia, hyperinsulinemia and glucose intolerance in our experiment. On the other hand, the oral administration of keishibukuryogan showed a certain beneficial effects by reducing the risk of type 2 diabetes, but it did not produce results comparable to those of pioglitazone.

One of the main risk factor for the progression of type 2 diabetes is the blood glucose level following an oral glucose load. In this study, we performed OGTT after 3 and 5 weeks of treatment. Treatment with keishibukuryogan for 3 weeks resulted in a tendency of lower blood glucose levels after oral glucose load, and with the prolongation of the administration period to 5 weeks, significant improvement was achieved. On the other hand, keishibukuryogan did not affect fasting serum glucose or insulin, or HOMA-IR.

Tumor necrosis factor- α (TNF- α) has been widely accepted as a crucial factor responsible for the development of impaired glucose metabolism and insulin resistance in obese and diabetic subjects [20–22]. It was demonstrated that infusion of TNF- α into rats resulted in insulin resistance in muscle [23], and that anti-TNF- α infusion in the animals resulted in an increase in glucose disposal [24]. Saghizadeh et al. also showed that the expression of TNF- α is higher in muscle of subjects with insulin resistance and type 2 diabetes [25]. Additionally, it is considered that the insulin resistant state of skeletal muscle is responsible for the rate of whole body glucose uptake, because skeletal muscle plays a major role in glucose disposal during glucose and insulin infusion [26,27]. Thus, in this study, TNF- α content in skeletal muscle was examined. The results showed a significant suppression of TNF- α protein content by pioglitazone, which is involved in the mechanisms of its anti-diabetic action [28]. Oral administration of keishibukuryogan

also resulted in significant suppression of TNF- α content in skeletal muscle of OLETF rats. It was reported that keishibukuryogan improved the microcirculation and protected endothelial function [11]. Further, according to this study, keishibukuryogan suppressed the secretion of TNF- α from adipose tissue. These effects are thought to improve the favorable effect on glucose intolerance. Therefore, it is suggested that keishibukuryogan has a favorable effect on impaired glucose metabolism in type 2 diabetes by improving glucose intolerance, and that some of this effect is derived from the reduction of TNF- α content in skeletal muscle.

Abnormalities of lipid metabolism are frequently complicated with type 2 diabetes [29–31]. Widen et al. reported that insulin-mediated glucose disposal was reduced in type 2 diabetic patients with hypertriglyceridemia compared with normotriglyceridemic patients [32]. In this study, we monitored the serum and hepatic lipid profiles. Our results confirmed the beneficial effects of pioglitazone against total cholesterol, triglyceride and free fatty acid levels in serum, and total cholesterol and triglyceride levels in liver, indicating the importance of the hypolipidemic properties of pioglitazone in the attenuation of impaired glucose metabolism and insulin resistance. Administration of keishibukuryogan for 3 weeks reduced serum total cholesterol and triglyceride levels with significance, and further reduction of these lipid parameters were observed at 6 weeks of treatment, whereas serum free fatty acid did not change during the experimental period. Additionally, it was observed that keishibukuryogan produced a hypolipidemic effect prior to attenuating impaired glucose tolerance. Previously, we demonstrated that keishibukuryogan had a hypolipidemic effect by reducing serum triglycerides in diabetic nephropathy rats [33]. From our present results and previous study, it was suggested that keishibukuryogan exerts a hypolipidemic effect in the diabetic body, and this effect contributes to amelioration of the insensitivity of peripheral tissues to insulin and glucose disposal, resulting in a reduction in glucose levels after oral glucose load. Further studies are required to clarify whether keishibukuryogan improves peripheral insulin sensitivity under the glucose and insulin infusion condition *in vivo*, and to reveal the correlation between lipid parameters and insulin resistance after keishibukuryogan treatment in greater detail.

Obesity, especially intra-abdominal visceral fat, is also frequently involved in type 2 diabetes, and adipose tissue-mass enlargement has been considered to be closely associated with insulin resistance [34]. Adipose tissue consists of adipocytes that take up glucose and store energy in the form of triglycerides. Recently, adipocytes were shown to be dynamic endocrine cells, which produce and secrete various bioactive molecules known as adipocytokines [35,36]. Of these adipocytokines, it was reported that higher plasma levels of leptin, plasminogen activator inhibitor-1, and TNF- α , and lower plasma levels of adiponectin accompanied obesity, and that dysregulation of adipocytokines participated in the development of insulin resistance [24,37–40]. Therefore, adipocyte hypertrophy and adipocytokines are thought to be key pathological contributors to insulin resistance, increasing the risk of type 2 diabetes mellitus. Our results showed that keishibukuryogan induces significant reduction in the serum leptin level after 6-week treatment, comparable to that

afforded by pioglitazone, whereas it had no effect on the serum adiponectin level. Also, we found that keishibukuryogan had a significant effect on adipocyte hypertrophy, as indicated by the decreased size of adipocytes in epididymal white adipose tissue. These observed effects of keishibukuryogan on the serum leptin level and white adipose tissue may add further support to the beneficial effect of keishibukuryogan against type 2 diabetes by reducing the risk associated with obesity.

Recently, the efficacy and usefulness of traditional herbal medicines are being reconsidered and reevaluated, because they have a wide range of therapeutic potential from their constitution of several herbs, and the risk of side effects is lower than that of modern Western medicines. In summary, our present results provide evidence that keishibukuryogan exerts improvement effects on impaired glucose tolerance, lipid metabolic parameters, TNF- α content in skeletal muscle, and adipocyte hypertrophy, which were characterized as insulin resistant conditions. These results suggest that keishibukuryogan has the potential to become a natural candidate with far-reaching effects for the protection and attenuation of type 2 diabetes, notwithstanding that further studies will be needed to establish the pharmacological mechanisms responsible for its effects.

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Conflict of interest

The authors state that they have no conflict of interest.

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